# zero_expert.py — technical

`src/moe_pruning/pruning/zero_expert.py · 34 lines · moe_pruning.pruning.zero_expert`

## Purpose

Parameter-free removed-expert placeholder.

## Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

### `class ZeroExpert`

lines 19-34 · bases nn.Module

Parameter-free placeholder for a removed expert (drop / reroute).

#### `ZeroExpert.forward`

```
def forward(self, hidden_states: torch.Tensor) -> torch.Tensor
```

lines 25-31 · raises RuntimeError

Return zeros of the expert output shape (error if must_be_unused and reached).

1. If `must_be_unused` and the input contains any token: raise `RuntimeError` (routing mask failure).
2. Otherwise return zeros of shape `(..., output_dim)` in the input's dtype/device.

## Errors raised

| Function | Exception | Message (f-string fields in braces) |
|---|---|---|
| `forward` | RuntimeError | {hidden_states.shape[:-1].numel()} token(s) reached a removed expert that should be unreachable (routing mask not applied?) |